

Subject: Re: ACA session web pages
From: Manfred Minimair <manfred@minimair.org>
Date: Tue, 05 May 2009 15:47:03 -0400
To: Gilles Picard <gilles.picard@etsmtl.ca>

Dear Gilles,
Please post the following three abstracts and the two attached ones in the chemistry session.

Best regards,

Manfred

Casey Henderson & Necmettin Yildirim

New College of Florida

Division of Natural Sciences

Ordinary differential equations are widely used to study dynamics of biochemical reaction networks. Possible regulatory mechanisms in these networks can be investigated by exploring how the system variables (concentration) respond to small changes in parameter values. Mathematically, this can be done by solving the model and sensitivity equations simultaneously. The number of sensitivity equations is a product of the number of variables and parameters. Even for simple networks, the derivations of these equations becomes challenging, if not impossible, when one wants to do it by hand. In this talk we will explain how to use Computer Algebra Systems, such as Maple, to perform a complete sensitivity analysis for biochemical reaction networks whose dynamics are described by a system of differential equations. In the second part of this talk, we will show how to apply our approach a model developed for the tryptophan operon of E.coli, which is a small genetic network responsible for production of tryptophan when it is not present in the environment.

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Fourier transforms in chemistry - diffraction and spectrometric applications
J. F. Ogilvie
Centre for Experimental and Constructive Mathematics, Department of Mathematics,
Simon Fraser University, 8888 University Drive, Burnaby, British Columbia V5A 1S6 Canada
and Escuela de Quimica, Universidad de Costa Rica, Ciudad Universitaria Rodrigo Facio, San Pedro de Montes de Oca, San Jose 11501 - 2060, Costa Rica

We employ Maple's ability to compute symbolic Fourier transforms of continuous curves to extract essential chemical information from interferograms produced with contemporary laboratory instruments. An interferogram is a function in the distance or time domain produced from the interference of waves, of which Fourier transformation into the wavenumber or frequency domain yields a distribution or spectrum that contains precise information about molecular structure and properties. We describe in turn each of four prototypical applications of continuous Fourier transforms - applied to diverse signals from coherent electron

scattering, coherent xray scattering,
microwave emission and infrared absorption - and then demonstrate how
the capabilities of software
for computer algebra enable the derivation of information in a
chemically meaningful form. We
distinguish between Fourier exponential, cosine and sine transforms that
yield novel complementary
information about molecular properties.

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Michael Anthony Whitehead, tony.whitehead@mcgill.ca
Department of Chemistry
McGill University, Montreal, Quebec, Canada

Classical Mechanical Methods for calculating Molecular Structure and
Quantum Mechanical methods for calculating Molecular Electronic
Properties are reviewed. Appropriate Theoretical Techniques are used to
investigate dendritic molecules containing silicon (Si) made from
3,5-dihydroxybenzyl alcohol (DBHA), and tin (Sn) dendrimers made from
1,3,5 tri ethynyl benzene (TEB) as well as purely organic dendrimers of
(TEB). Nanotubes formed from SMA (poly (styrene alt.maleic anhydride)
[3] and from an imide derivative (SMI) [4] are considered. Finally the
effect of water on the calcium carbonate CaCO₃ Aragonite crystal (100)
face is considered and contrasted with the effect of water on the (10.4)
face of CaCO₃ Calcite crystal.

Gilles Picard wrote:

Dear Manfred,

I produced basic webpages for your 2 sessions:
<http://aca2009.etsmtl.ca/sessions/elimination.html>
<http://aca2009.etsmtl.ca/sessions/chemistry.html>

As you can see, these are now online. I did put a deadline for
submission of talks, May 29. Some of the other webpages have this deadline.
I can change the date, or remove it or edit easily these pages, so do
not hesitate to ask me for corrections or to add information.
When you have accepted talks and abstract, just email them to me and I
will put them on your webpage.

Best regards,
Gilles

Manfred Minimair wrote:

Dear Gilles,
Please set up web pages for sessions 4 and 5 as you did for 11.

Best regards,

Manfred

The original MIME headers for this attachment are:

Content-Type: text/x-tex;
name="abstract_of_chung-feller_I[1].tex"
Content-Transfer-Encoding: 7bit
Content-Disposition: inline;
filename="abstract_of_chung-feller_I[1].tex"

abstract_of_chung-feller_I.pdf	Content-Type: application/pdf
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abstract_of_chung-feller_I[1].tex	Content-Type: text/x-tex
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Computer Algebra Application in Chemical Spectroscopy.doc	Content-Type: application/msword
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